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MANAGEMENT SCIENCE
Vol. 64, No. 7, July 2018, pp. 3365–3379
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Temporary Incentives Change Daily Routines: Evidence from a Field Experiment on Singapore's Subways

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Received: January 7, 2016
Revised: May 31, 2016; August 12, 2016
Accepted: September 22, 2016
Published Online in Articles in Advance: May 9, 2017

https://doi.org/10.1287/mnsc.2017.2731

Abstract. We show that temporary incentives can lead to lasting changes in people's daily routines through a field experiment. In randomized controlled trials on Singapore's subways, commuters in the treatment group received a full rebate of their fares for 10 weeks if they exited fare gates before the start of morning peak-travel time. The objective of the fare promotion was to alleviate peak-time congestion by encouraging commuters to travel prepeak. This promotion resembles an actual public program implemented in Singapore in 2013. Using participants' fare cards, we obtained a rich data set for 60 weeks' worth of trips. A difference-in-differences analysis reveals that the effect of the free-trip offer was modest but robust during the promotional period: on average, trips were 5.5% more likely to end before the morning peak time. More encouragingly, the modest shift toward early travel strongly persisted even seven months after the end of the free-trip offer. This seemingly small shift could imply large financial savings on infrastructure investment. We find suggestive evidence that travel regularity and consistency during the promotion appear to be important for forging persistence.

History: Accepted by Uri Gneezy, behavioral economics.
Funding: N. Yang gratefully acknowledges financial support from the National University of Singapore Humanities and Social Sciences (HSS) Seed Fund [R-313-000-112-646] and the Land Transport Authority of Singapore's Land Transport Innovation Fund.
Supplemental Material: Data and the online appendix are available at https://doi.org/10.1287/mnsc.2017.2731.

Keywords: subway congestion • price promotion • persistence in habitual behavior • field experiment

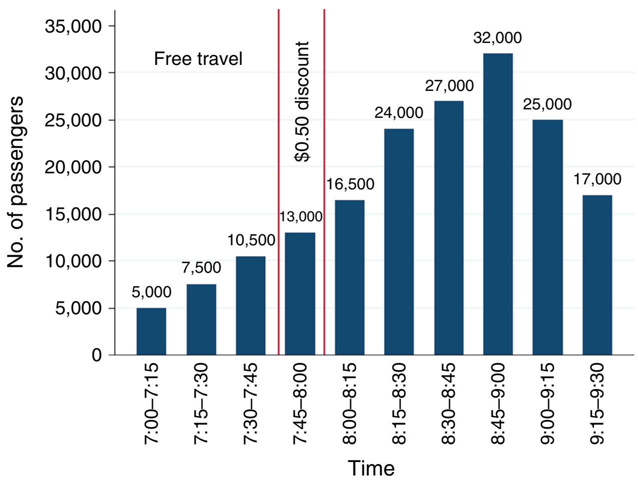
1. Introduction

In recent years, "nudging" people into favorable habitual behavior by providing short-term financial incentives has become increasingly popular. For instance, new provisions in the Affordable Care Act have motivated the establishment of employee wellness programs that give short-term rewards for healthy behavior (Madison 2015), and Singapore's Health Promotion Board launched a "National Steps Challenge" that gave out shopping vouchers for accumulating walking steps within a half-year window.[1] Practices outside of the health domain, however, have been relatively scarce. In this study, we report some success in influencing people's daily routines for better social outcomes from a field experiment in the context of peak-time urban transit-demand management.

In metropolitan areas such as New York, Beijing, Tokyo, and Singapore, as many as four or five passengers can be crowded into one square meter of standing space during peak morning hours. Not only is this a miserable experience for passengers, but crowded trains also discourage drivers from switching to public transit and therefore contribute to congestion on roads. To accommodate the peak-time demand surge, transit authorities invest heavily in personnel, fleet, and infrastructure, which are often idle during nonpeak times. To reduce such wasteful investment, several transit authorities around the globe have experimented with differentiated fare schemes to smooth demand during peak times. Typically, this involves a fare surcharge or temporary nullity of discounts during peak hours. Cervero (1986) reviews a large number of such practices and concludes that peak-time commuters are fairly insensitive to surcharges. In addition, surcharges add to peak-time commuters' discomfort, generate resentment among frequent customers, and are deemed by transit authority practitioners to be "politically unpopular" (Henn et al. 2010).

We set up a field experiment in Singapore to examine the effectiveness of temporarily offering free prepeak trips as a more welcoming differential-fare scheme. We recruited more than 350 regular morning peak-time subway commuters to participate in the experiment and randomized them into one control and two treatment groups. For 10 weeks, one *early treatment group* was allowed to travel free of charge if they exited the fare gates before 7:45 A.M. and the other *late treatment group* before 8:00 A.M. To avoid disappointing those who narrowly missed the free travel cutoff time, we also offered a 50-cent (Singaporean) discount during

Figure 1. (Color online) Precampaign Daily Average Number of Commuters Exiting the 16 City-Center Stations on Weekdays



Source. LTA's campaign website http://www.lta.gov.sg/data/apps/news/press/2013/20130416_FreeTravel_Annex.pdf (accessed January 1, 2016). Data are from the first two weeks of March 2013 and exclude school holidays.

a 15-minute grace period from 7:45 to 8:00 A.M. and 8:00 to 8:15 A.M. to the respective treatment groups. The control group always paid the normal fare, which did not depend on exit time.

In Singapore, governmental economic interventions on transit are not uncommon,[2] and our experimental design resembled an actual public program. In June 2013, as part of a package plan to alleviate subway crowding, Singapore's Land Transport Authority (LTA) launched a campaign, "Travel Early, Travel Free," that offered free trips to those who exited from one of the 16 (extended to 18 following the completion of a new subway line in 2014) city-center subway stations before 7:45 A.M. on weekdays. Commuters who exited within the next 15 minutes received a 50-cent discount. Figure 1 shows the daily average number of exits at the 16 city-center stations on weekdays before the campaign and the promotion's cutoff times. The campaign's objective was to smooth the surged crowd during the peak window from 8:30 A.M. to 9:00 A.M. The campaign was still in place as of May 2016; preliminary data indicate a reduction of 7%–8% in the number of commuters during the morning peak window.[3] The Singaporean case is not a unique one: Hong Kong and Beijing, which also have subway-congestion problems, both adopted similar prepeak discount schemes, instead of peak-time surcharges, for one-year trials in 2015.[4]

Nonetheless, there are two important reasons to conduct the policy evaluation in our experiment. First, we wish to examine whether temporary monetary incentives' impact on daily routines persists after their removal. With a projected annual expense of more than S$15 million (Land Transport Authority 2013) (S$ stands for Singapore dollars; 1 U.S. dollar ≈ 1.4 Singapore dollars), the "Travel Early, Travel Free" campaign was designed to be a time-limited trial. Therefore, its lasting effect is of great policy relevance. Second, our experiment's randomized controlled trials design allows for a straightforward difference-in-differences analysis, which disentangles the "Travel Early, Travel Free" promotional fare's effect from confounding measures in the package plan, such as more frequent trains, targeted advertising, and incentives for employers to allow flexible work schedules. Our experiment ran concurrently with the "Travel Early, Travel Free" campaign, but at different stations.

Most Singaporean commuters, including all subjects in our experiment, use smart cards to pay subway fares. Before boarding a train, they tap the card at the station of origin to open the fare gates. Distance-dependent fares are automatically collected when they again tap the card before exiting the destination station. For each subway ride, the smart card records the origin, destination, start and finish times, and fare. This fare-collection system greatly facilitated the implementation of our experiment and minimized measurement error in the data. To the best of our knowledge, previous studies of public-transit pricing (e.g., De Witte et al. 2006, Paulley et al. 2006, Sharaby and Shiftan 2012) have not used this data-collection technique.

We collected 60 weeks' worth of data from participants' cards for roughly 100,000 subway rides before, during, and after the promotional-fare period and found a modest but robust impact of the promotional fare on shifting commuters' travel times from peak to prepeak hours during the promotional period. A typical morning trip made by commuters in the treatment groups was 2% more likely to end by 7:45 A.M. and 5.5% more likely to end by 8:00 A.M. These seemingly small shifts could be equivalent to an expensive injection of additional train capacity to the network during the peak period. Even more encouragingly, the impact strongly persisted over the next seven months with only small decay, leading to vast improvement of the fare scheme's cost effectiveness.

In a small but growing literature, researchers have documented persistence or its absence following incentive programs designed to alter habitual behavior. In a series of studies in which subjects were paid to incentivize gym attendance, both Charness and Gneezy (2009) and Acland and Levy (2015) find that subjects maintained their increased attendance levels months after the incentive ended. Royer et al. (2015) followed their subjects for several years and detected persistence in exercising for those who had voluntarily signed a commitment contract. Loewenstein et al. (2016) show strong persistence among elementary school children in eating healthy after receiving incentives for doing so over a number of weeks. In a review article, Gneezy et al. (2011) discuss several examples in which the treatment effect disappears shortly after the incentive is discontinued, especially in the context of smoking cessation and weight loss. In transportation research, both Fujii and Kitamura (2003) and Thøgersen (2009) find that offering one month's worth of free bus tickets has a long-lasting but diminishing effect on reducing automobile use. Analysis of our rich experimental data presents suggestive evidence that travel regularity and consistence during the promotion appear to be important for forging persistence.

Recent research by Anderson (2014) on the case of Los Angeles suggests that keeping peak-time commuters on public transit is likely to have a large impact on roadway congestion, which justifies the cost of subsidizing public transit. However, Basso et al. (2011) and Basso and Silva (2014) argue that the marginal effect of subsidies on attracting commuters to public transit diminishes rapidly. Our experiment weighs in on this policy discussion by showcasing the role of behavioral factors, such as habit formation, in optimizing public-transit subsidies. More broadly, our findings also connect to economic and behavioral interventions (e.g., those reviewed in Allcott and Mullainathan 2010) that aim to ease congestion in the energy market, where the similar, lumpy nature of infrastructure investment and vast user bases also implies that a small behavioral shift toward energy conservation can have a profound impact on financial and environmental costs.

2. The Field Experiment

In Singapore's subway system, Mass Rapid Transit (MRT), about 99.5%[5] of trips are paid using a contactless smart card, which facilitated our experiment in three ways. First, although most of the cards are anonymous,[6] each has a unique 16-digit serial number. With this number in hand, one can extract the owner's travel records from a central database. Second, on both entering and exiting a subway station, riders are required to tap the card on a card reader, which allowed us to precisely identify promotion-eligible trips. Third, the card also serves as a cashless payment method for many diners, shops, and taxicabs, and therefore its stored value is a close substitute for money. This allows us to pay all experiment participants by crediting their accounts electronically. Unfortunately, because the card is anonymous, it cannot be tied to individual characteristics or contact information—and even if it could, using such information without prior consent would be a violation of participants' privacy. Therefore, we had to recruit commuters to volunteer for this experiment who were willing to provide their cards serial number and contact information.

Our experiment ran concurrently with the "Travel Early, Travel Free" campaign, but at different locations: We conducted recruitment on the 11 weekdays between August 28 and September 11, 2014, at two busy subway stations outside the city center that were not included in the ongoing "Travel Early, Travel Free" campaign. Recruitment teams, which consisted of undergraduate students from the National University of Singapore, distributed information pamphlets to commuters exiting these stations and registered on-site sign-ups. To minimize contact with commuters whose travel windows, without any intervention, fell outside the peak hours, recruitment teams solicited volunteers only between 8:00 A.M. and 9:00 A.M. The information pamphlets (see the online appendix) stated that participants might be chosen to enjoy promotional subway fares in September and October. To enroll, a commuter provided the 16-digit serial number of the travel card used, his or her email address, and, optionally, his or her mobile phone number. Each commuter who registered on-site was given a package of cookies as a token of appreciation. The information pamphlets stated that commuters also could sign up by sending a text message or email or by completing an online registration form. In total, 508 commuters signed up on-site and 339 did so off-site.

For all 847 registered commuters, we backtracked their travel histories from August 28, the start of recruitment, to three weeks earlier. We then selected "frequent" commuters by identifying those whose weekday trips' exit times fell between 7:45 A.M. and 9:15 A.M. at least 3 out of the possible 5 times within one week, 5 times out of 10 within two weeks, or 7 times out of 15 within the three weeks before August 28. After screening, 379 commuters qualified. We further randomized participants into one control and two treatment groups. The early treatment group (ET) would receive a full rebate of their subway fare if they exited the station before 7:45:00 A.M. on a weekday and a 50-cent rebate if they exited between 7:45:01 A.M. and 8:00:00 A.M. The late treatment group (LT) would receive the full rebate if their weekday exit occurred before 8:00:00 A.M. and a 50-cent rebate if the exit occurred between 8:00:01 A.M. and 8:15:00 A.M. ET and LT differed only in cutoff times. We designed the two treatments with the goal of calibrating an effective policy rather than testing alternative theories. The control group (CG) would receive no promotional fare and therefore paid the full fare at all times.

Immediately after concluding recruitment on September 11, we began the intervention by sending all 379 qualified commuters an optional online survey via email about their travel preferences and demographics. We took our best guess for a number of unrecognizable handwritten on-site registration email addresses, which resulted in 31 incorrect addresses. This further reduced the number of participants to 348, of whom 120 were ET, 109 LT, and 119 CG. Hereafter, they are referred to as the "subjects." They were given until September 22, the start of the promotional fare period, to fill out the optional online survey for a S$10 reward. One hundred sixty-three commuters completed the survey. For those in the CG, this was our last contact with them. For those in the ET and LT, the last screen of the online survey explained the details of the promotional fare the subject would receive over the next 10 weeks. Because the survey was not compulsory, we also sent the same information about the promotional fare via email (twice) and a brief version via text message (once). Despite our best efforts to notify subjects, we acknowledge the possibility that some subjects might have overlooked our messages, which would introduce a bias toward insignificant results. We believe that such negligence, if any, is of practical relevance: no matter an advertising campaign's intensity, any actual fare promotion could still be neglected by a fraction of the general public. If we had mandated an affirmative reply for a subject to be included in one of the treatment groups, we would likely have missed some subjects who were not attracted by the promotional fare and thereby generated a bias toward significance.

The 10-week promotion started on September 22 and ended on November 28. Beginning in week 2, all ET and LT subjects received weekly reminder emails about how much they had accrued in rebates (1) during the previous week and (2) overall. These emails included the statement, "Please continue to ride MRT at a time that you think is best for you." After the fare promotion's conclusion, we sent an exit survey to all ET and LT subjects, which came with a S$20 reward for completion. One hundred twenty-four commuters finished the survey. We paid all rebates and survey rewards in the form of value added to travel cards at three points: week 5 and week 9 during the promotional period, and two weeks after the promotion ended. After the exit survey, we continued to monitor all subjects' travel records until June 26, 2015, which was roughly seven months after the fare promotion ended. During this period, we had no more contact with the subjects. On December 10, 2015, four subjects attended a focus-group discussion session for another research project of ours and were asked a few questions related to this experiment. See the online appendix for results from both surveys and the focus-group discussion.

3. Results

3.1. Data and Descriptive Results

We extracted up to 60 weeks' worth of travel records from each subject's travel card, including 20 weeks before, 10 weeks during, and 30 weeks after the promotional fare offering. We then examined all subway trips on weekdays (Monday through Friday, excluding public holidays) that involved a treatment station as an entry or exit point, which meant that subjects' home-bound commutes were recorded as well. In total, we obtained data for 96,635 trips, of which 50,183 ended before 9:30 A.M. Hereafter, we refer to these as "morning trips."

Before proceeding to our analysis of the treatment effect, we report in Table 1 that our random assignment of subjects does create balanced treatment and control groups in terms of travel patterns and demographics. To better control for commuters' individual heterogeneity, we use individual fixed effects instead

Table 1. Summary Statistics and Sample Balance

	Control group	Early treatment	Late treatment
Age	36.1	36.8	35.8
Female (%)	58.30	59.80	67.90
Self-report to have flexible work hours (%)	43.80	41	34.50
Income (median)	4K–5K	3K–4K	3K–4K
Family income (median)	5K–8K	4K–5K	4K–5K
Presurvey response rate (%)	40.34	50.83	49.54
Postsurvey response rate (%)	NA	51.67	56.88
Prepromotion personal average exit time	8:30	8:26	8:28
Prepromotion personal average probability of ending trip by 7:45 A.M. (%)	3.21	3.08	3.51
Prepromotion personal average probability of ending trip by 8:00 A.M. (%)	9.37	8.12	9.06
Prepromotion personal average probability of ending trip by 8:15 A.M. (%)	25.89	28.73	25.53
No. of trips per person prepromotion	55.15	61.09*	56.80
No. of trips per person during promotion	26.78	28.74	27.59
No. of trips per person postpromotion	66.12	76.39*	70.98
Average weekly trips per person prepromotion	3.51	3.68	3.59
Average weekly trips per person during promotion	2.93	3.00	2.95
Average weekly trips per person postpromotion	3.03	3.22	3.00
Personal average before-discount subway fare prepromotion	1.35	1.32	1.33
Personal average before-discount subway fare during promotion	1.35	1.33	1.33
Personal average before-discount subway fare postpromotion	1.37	1.34	1.31
Attrition rate (%)	50.42	44.17	48.62
Sample size	119	120	109

Note. Pairwise t-test results for equal means (ET versus CG and LT versus CG), except for income and family income.
$^{*}p < 0.1$.

of group fixed effects in our difference-in-differences analysis.

Two statistics in Table 1 are worth commenting on. First, public transport fares are considerably cheaper in Singapore than in comparable developed countries. Our subjects' average subway fare without promotion is only about S$1.3. The savings for a full month's (20 workdays) worth of the average fare accounts for less than 1% of their median income. This suggests that our promotional fare only offers a small monetary incentive. Second, the attrition rate, defined as the percentage of subjects who did not have any travel records in the last 30 days of the observation period, is notably high. Attrition is most likely due to subjects' changing to a new or different travel card without notifying us, and it causes our panel data to be unbalanced—yet it biases our results only if it is selective on treatment-related factors. Evidence from both discrete-choice models and hazard models (see Table 2) appears to suggest otherwise.

The upper left panel of Figure 2 shows the weekly average exit time for morning trips for CG, ET, and LT. The week the promotion started is designated week 1 in the figure. It appears that exit times for both ET and LT drop immediately in response to the fare promotion and drive a wedge between them and CG exit time. Although exit times for both treatment groups pick up again after the promotion ends, the difference between them and CG exit time persists in the postpromotion period. In the upper panel of Figure 3, we show the kernel densities of exit times for the control and treatment groups before, during, and after the promotional period. The promotion shifts the treatment groups' distribution to the left of the control group's, and the difference remains postpromotion. We further use Q-Q plots in the lower panels of Figure 3 to compare the distribution functions. These plots show that the distribution functions are similar prepromotion, whereas for many of the given quantiles before 8:30 a.m. during and after promotion, the control group's exit time is larger (later) than that of the two treatment groups.

It is worth noting that the magnitude of the exit time changes is economically small; one might question whether a five-minute shift in exit time helps to ease peak-time crowdedness. Indeed, for a shift of such

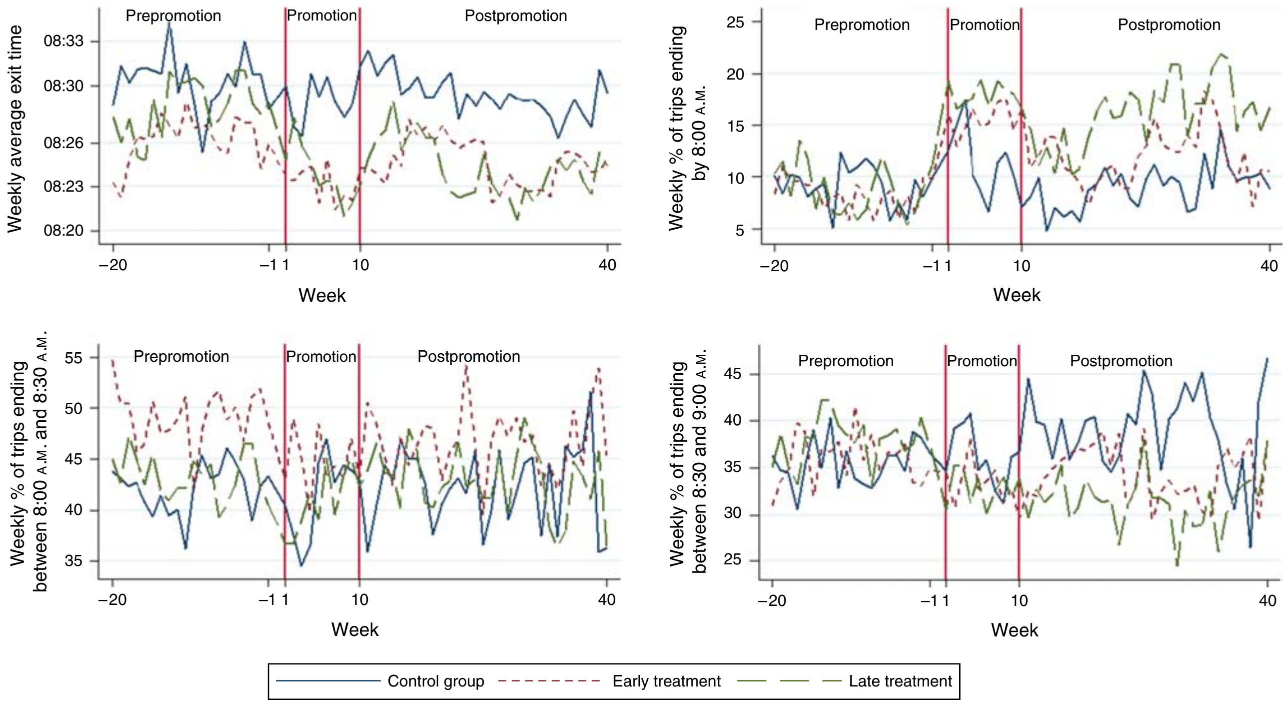
Table 2. Discrete-Choice Model and Hazard Model on Attrition

Variables	(1) Probit	(2) Logit	(3) Probit	(4) Logit	(5) Exponential	(6) Exponential
Number of times receiving full rebate	0.036	0.056	0.047	0.080	−0.034	−0.031
	(0.029)	(0.046)	(0.050)	(0.082)	(0.034)	(0.048)
Total rebate received	−0.015	−0.024	−0.011	−0.019	0.016	0.008
	(0.019)	(0.030)	(0.033)	(0.053)	(0.021)	(0.035)
Number of times receiving 50-cent rebate	0.019	0.031	0.006	0.011	−0.020	−0.012
	(0.016)	(0.026)	(0.025)	(0.041)	(0.018)	(0.026)
Having 5 *or more consecutive gaps during promotion*	−0.066	−0.107	−0.582**	−0.957**	−0.083	0.124
	(0.187)	(0.299)	(0.263)	(0.432)	(0.207)	(0.314)
Prepromotion personal average probability to end trip by 8:00 a.m.	−0.529	−0.855	−2.156*	−3.525*	−0.075	0.676
	(0.619)	(1.013)	(1.246)	(2.076)	(0.656)	(1.049)
Standard deviation of exit time prepromotion	−0.004	−0.007	0.024	0.039	0.004	−0.003
	(0.011)	(0.017)	(0.018)	(0.029)	(0.011)	(0.016)
Average exit time prepromotion (lapsed minutes since midnight)	0.001	0.002	−0.003	−0.005	−0.002	0.004
	(0.005)	(0.009)	(0.009)	(0.014)	(0.006)	(0.009)
Had at least one trip that ended by 8:00 *prepromotion*	0.220	0.354	0.335	0.540	0.009	0.095
	(0.174)	(0.280)	(0.265)	(0.433)	(0.185)	(0.283)
Age			0.004	0.005		0.007
			(0.011)	(0.018)		(0.010)
Female			0.056	0.088		0.039
			(0.227)	(0.372)		(0.247)
Income			−0.127*	−0.204*		0.086
			(0.076)	(0.123)		(0.075)
Flexibility			0.181	0.297		−0.187
			(0.240)	(0.397)		(0.244)
Completed at least one survey	−0.109	−0.175			0.168	
	(0.163)	(0.263)			(0.176)	
Observations	348	348	163	163	268	127

Notes. Columns (1)–(4) are discrete-choice models with attrition (= 1) as the dependent variable. Columns (5) and (6) are hazard models with the spell of being observed as the dependent variable and an exponential hazard function. Standard errors are in parentheses.

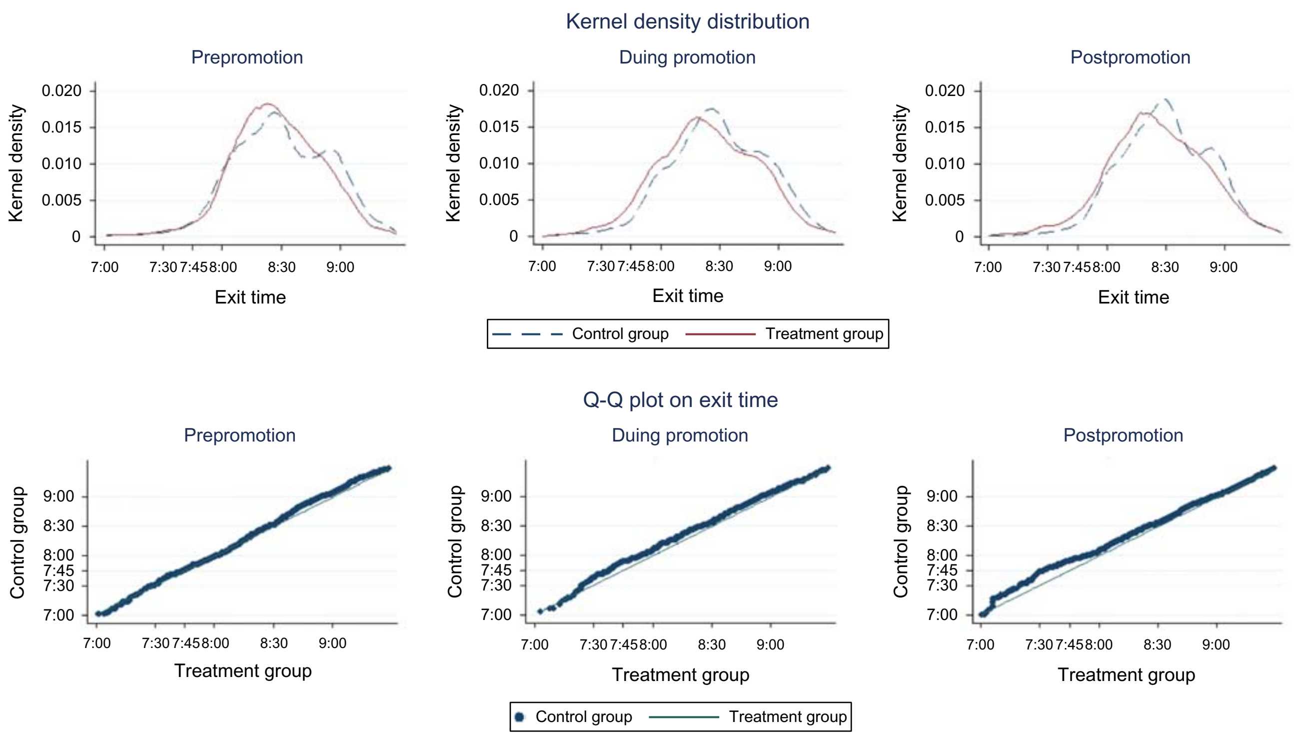
$^{**}p < 0.05$; $^{*}p < 0.1$.

Figure 2. (Color online) Trends in Trips' Exit Time: Prepromotion, During Promotion, and Postpromotion



a small size to have an impact, it would have to effectively move subjects from peak time to prepeak time. The upper right panel of Figure 2 shows the weekly average percentage of morning trips that ended before 8:00 A.M., the cutoff for ET to receive a 50-cent discount and for LT to receive the full rebate. In the pretreatment weeks, only 5%–10% of morning trips ended before 8 A.M. across all three subject groups. This percentage rises to and remains at more than 15% during the promotional period for both ET and LT. Despite the drops

Figure 3. (Color online) Kernel Densities and Q-Q Plots for Control and Treatment Groups' Exit Time: Prepromotion, During Promotion, and Postpromotion



at the end of the promotion,[7] the postpromotion percentages for ET and LT are consistently higher than those for CG. The lower left panel shows that the percentage of trips ending between 8:00 A.M. and 8:30 A.M. seems largely unchanged, while the lower right panel shows a reduction in the percentage of trips ending during the peak window of 8:30 A.M.–9:00 A.M. for ET and LT, which is more pronounced postpromotion than during promotion. Since the upper left panel tells us that on average, subjects in the treatment groups only shifted their trips by a few minutes, we hypothesize that some of them shifted from the 8:00 A.M.–8:30 A.M. window to before 8:00 A.M., while some of them shifted from the 8:30 A.M.–9:00 A.M. window to 8:00 A.M.–8:30 A.M.

3.2. Effect During and After the Promotional Fare

To examine whether the plotted differences are statistically significant after controlling for commuter and trip heterogeneity, we estimate two trip-level linear difference-in-differences models. Because the ET and LT only slightly differ in the promotion's cutoff times, we examine the average effect of the two treatments in the first model and then, in the second model, the difference between the treatments. The first model is

$$dv_{ij} = \beta_1 During_{ij} + \beta_2 During_{ij} \times Trt_i + \beta_3 Post_{ij} + \beta_4 Post_{ij} \times Trt_i + \beta_5 X_{ij} + \nu_i + \varepsilon_{ij}. \quad (1)$$

In this regression equation, dv_{ij} is the dependent variable associated with subject i's morning trip j. We use seven different dependent variables in our analysis: the morning trip's exit time (as the number of lapsed minutes since midnight) and six binary indicators for whether the morning trip ends before 7:45 A.M., before 8:00 A.M., before 8:15 A.M., between 8:15 A.M. and 8:30 A.M., between 8:30 A.M. and 9:00 A.M., and between 9:00 A.M. and 9:30 A.M.. Independent binary variables $During_{ij}$ and $Post_{ij}$ indicate whether trip j is during or after the promotional period. Trt_i is another binary indicator and has a value of 1 if subject i is in ET or LT and 0 if he or she is in CG. Coefficients for the interaction terms measure the promotional fare's treatment effects during and after the promotional period. We collect three trip-level characteristics in X_{ij} as controls: the before-discount trip fare (which is proportional to the distance traveled), a binary indicator for whether the trip is connected with a bus transfer, and dummies for the day of the week the trip is made. We further include individual fixed effects ν_i to flexibly control for commuter heterogeneity. This means that we do not include dummy variables for whether subject i belongs to a treatment group or his or her demographics. Instead, we have shown in Table 1 that there is no systematic difference in travel patterns and demographics between treatment and control groups. Standard errors are clustered at the subject level (Bertrand et al. 2004).

Table 3 shows the estimation results for the aforementioned seven dependent variables. During the promotional period, the small shifts in minutes of exit time (regression (1)) generally do make an average trip moderately but significantly more likely to end by the promotion's cutoff time (regressions (2)–(4)). Specifically, an average trip made by subjects from either ET or LT is 2%, 5.5%, and 4.2% more likely to end before 7:45 A.M., 8:00 A.M., and 8:15 A.M., respectively. In the postpromotional period, these treatment effects largely persist with small decays. An average trip for ET or LT is still 2.1% and 4% more likely to end before 7:45 A.M. and 8:00 A.M., respectively. Both estimates are statistically significant.

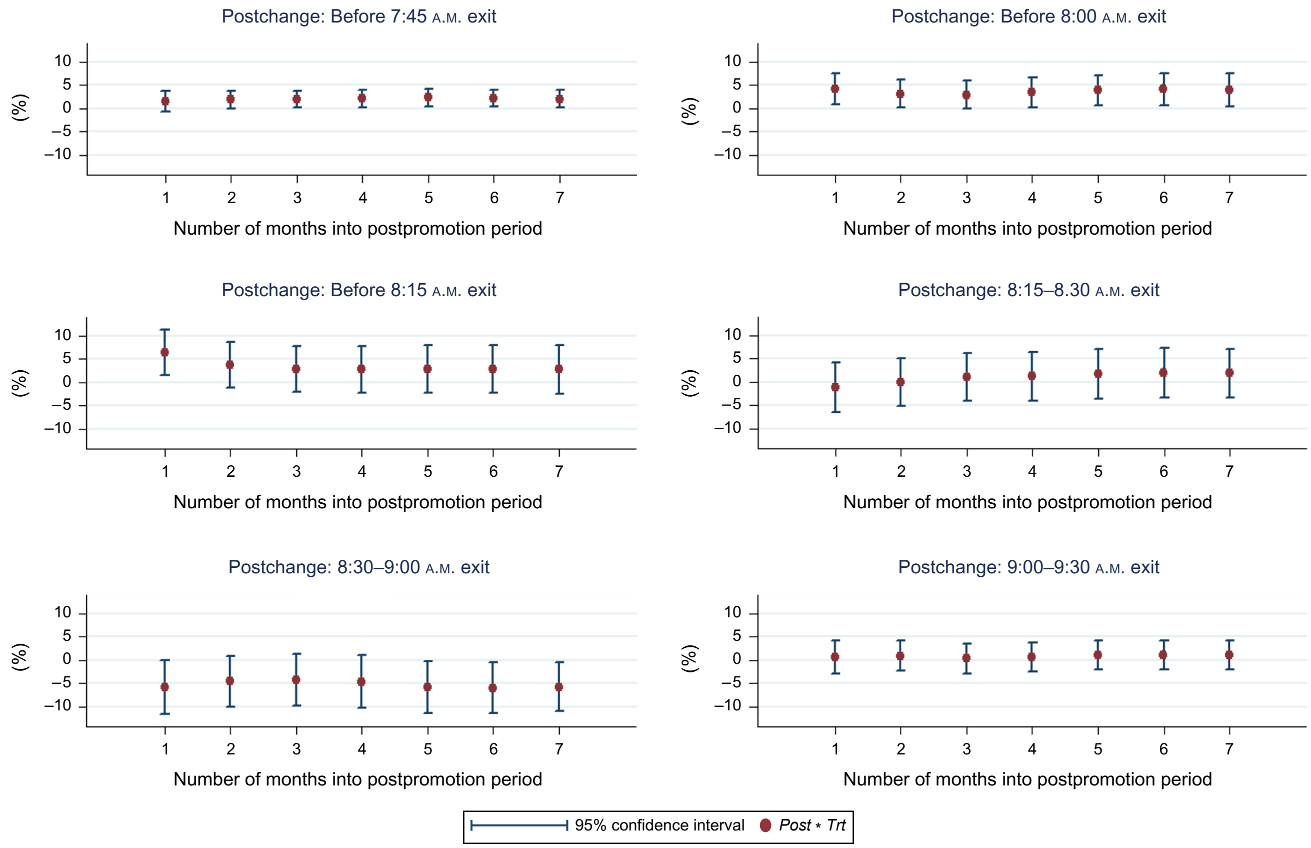
Table 3. Trip-Level Difference-in-Differences Estimates: Treatments Combined

Variables	(1) Exit time (min)	(2) Before 7:45	(3) Before 8:00	(4) Before 8:15	(5) Bet. 8:15 and 8:30	(6) Bet. 8:30 and 9:00	(7) Bet. 9:00 and 9:30
During	−1.213	−0.000	0.017	0.013	0.012	−0.013	−0.011
	(0.763)	(0.006)	(0.012)	(0.015)	(0.016)	(0.018)	(0.011)
During × *Trt*	−1.836*	0.020**	0.055***	0.042**	−0.039**	−0.003	−0.001
	(1.031)	(0.009)	(0.019)	(0.021)	(0.020)	(0.023)	(0.013)
Post	1.189	−0.005	−0.006	−0.013	−0.024	0.042*	−0.006
	(0.893)	(0.006)	(0.012)	(0.021)	(0.023)	(0.022)	(0.013)
Post × *Trt*	−2.278*	0.021**	0.040**	0.028	0.019	−0.058**	0.011
	(1.207)	(0.009)	(0.018)	(0.026)	(0.027)	(0.027)	(0.015)
Observations	50,183	50,183	50,183	50,183	50,183	50,183	50,183
R-squared	0.594	0.271	0.365	0.530	0.309	0.416	0.343
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trip-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Trip-level controls include the before-discount trip fare, a binary indicator for whether the trip is connected to a bus transfer, and a dummy for day of week when the trip is made. Robust standard errors are in parentheses (clustered at the individual level).

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Figure 4. (Color online) Postpromotion Treatment Effect, by Months After Promotion



For any morning trip, which is defined as a subway ride that ends before 9:30 A.M., the four dependent variables in regressions (4)–(7) complete the possibilities for when it ends. The coefficient estimates in these four regressions suggest that during the promotional period, subjects in ET and LT shifted trips primarily out of the early peak window 8:15 A.M.–8:30 A.M. and into before 8:15 A.M. Changes in the probability that trips will end after 8:30 A.M. are estimated to be small and statistically insignificant. Such patterns suggest that subjects were making relatively small shifts to take advantage of the monetary rewards. In the postpromotional period, CG subjects appeared to be 4.2% more likely to end a trip during the peak window 8:30 A.M.–9:00 A.M., possibly because several measures, including additional train service, improved peak travel comfort during the postpromotional period. However, treatment-group subjects still attempted to avoid peak hours in the absence of monetary incentives: we find a statistically significant shift of 5.8% trips out of the peak window, 8:30 A.M.–9:00 A.M. These trips were spread across the other three time windows, though the probabilities of shifting into these windows are not precisely estimated. To examine whether the postpromotion treatment effect wanes over time, we estimate model (1) using data including 1, 2, . . . , 7 months after the end of promotion. Figure 4 plots the coefficient estimates for $Post_{ij} \times Trt_i$ and their 95% confidence intervals. Apart from a small decay after the first month in some cases, the effect is largely stable months into the postpromotion period. These patterns give us confidence that the effect is likely to persist beyond the observation window.

To explore the difference between the two treatments, we further estimate the following model:

$$\begin{aligned} dv_{ij} = {} & \beta_1 During_{ij} + \beta_2 During_{ij} \times Trt_i + \beta_3 During_{ij} \times ET_i \\ & + \beta_4 Post_{ij} + \beta_5 Post_{ij} \times Trt_i + \beta_6 Post_{ij} \times ET_i \\ & + \beta_7 X_{ij} + \nu_i + \varepsilon_{ij}, \end{aligned} \tag{2}$$

which differs from (1) in additional regressor ET_i, a binary variable indicating whether subject i belongs to ET ($ET_i = 1$) or not ($ET_i = 0$). Under this specification, β_2 and β_5 capture the during-promotion and postpromotion treatment effect of LT, while β_3 and β_6 measure the during-promotion and postpromotion differences between the treatment effects of ET and LT.

Table 4 presents the estimation results of (2) for the same seven dependent variables. It shows that the two treatments' effects in increasing prepeak trip probability are similar in magnitude during the promotional period. The postpromotion treatment effects are somewhat different. For instance, in regression (6), the point

Table 4. Trip-Level Difference-in-Differences Estimates: Treatments' Difference

Variables	(1) Exit time (min)	(2) Before 7:45	(3) Before 8:00	(4) Before 8:15	(5) Bet. 8:15 and 8:30	(6) Bet. 8:30 and 9:00	(7) Bet. 9:00 and 9:30
During	−1.213	−0.000	0.017	0.013	0.012	−0.013	−0.011
	(0.763)	(0.006)	(0.012)	(0.015)	(0.016)	(0.018)	(0.011)
$During \times LT$	−2.260*	0.019*	0.056**	0.042*	−0.022	−0.021	0.001
	(1.220)	(0.011)	(0.022)	(0.023)	(0.023)	(0.027)	(0.017)
$During \times (ET - LT)$	0.780	0.002	−0.001	0.000	−0.033	0.033	−0.003
	(1.388)	(0.013)	(0.028)	(0.029)	(0.023)	(0.027)	(0.014)
Post	1.188	−0.005	−0.006	−0.013	−0.024	0.042*	−0.006
	(0.893)	(0.006)	(0.012)	(0.021)	(0.023)	(0.022)	(0.013)
$Post \times LT$	−2.435	0.032**	0.055**	0.031	0.023	−0.071**	0.017
	(1.645)	(0.015)	(0.025)	(0.036)	(0.033)	(0.033)	(0.019)
$Post \times (ET - LT)$	0.279	−0.021	−0.027	−0.005	−0.007	0.023	−0.011
	(1.699)	(0.016)	(0.027)	(0.033)	(0.030)	(0.031)	(0.018)
Observations	50,183	50,183	50,183	50,183	50,183	50,183	50,183
R-squared	0.594	0.272	0.365	0.530	0.309	0.417	0.343
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trip-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Trip-level controls include the before-discount trip fare, a binary indicator for whether the trip is connected to a bus transfer, and a dummy for the day of week when the trip is made. Robust standard errors are in parentheses (clustered at the individual level).
$^{**}p < 0.05$; $^{*}p < 0.1$.

estimates for diverging trips out of the 8:30–9:00 A.M. window indicate a 7.1% reduction for LT and only 4.8% for ET. Nonetheless, none of the estimates for the difference between LT and ET, during promotion or postpromotion, is statistically significant. We henceforth pool the treatments for a more concise presentation and repeat all of the analyses with separated treatments in the online appendix.

Some back-of-the-envelope calculation helps to put the seemingly small percentages in Tables 3 and 4 in perspective: the vehicle fleet for two of Singapore's busiest subway lines currently has 141 trains, with another 57 on order. Of the entire fleet, at least 90% are expected to be deployed during peak morning hours. The LTA's objective is to have more than 180 trains running concurrently during peak morning hours on these two lines by 2019. The 5.8% reduction in trips during the peak window implies that 5.8% of the peak-period train capacity is freed up for a less crowded travel experience. This is equivalent to injecting an additional S$140 million[8] worth of trains on those two lines alone, or 2.3% of the fiscal budget allocated to Singapore's transport development in 2014.

To compare the cost effectiveness of the treatments versus purchasing more trains, we need to get an estimate of the number of trips shifted from one time window to another, or the *intensive margin* of the treatment effect. To this end, we first need to examine whether the treatments induce commuters to make additional trips, or the *extensive margin*. A positive extensive margin would be undesirable from the public-policy standpoint, as the treatment-induced trips contribute to crowdedness even if they occur during prepeak times. Because the promotions have rather early cutoff times and small monetary rewards, we expect the extensive margin to be very small. Three regressions in Table 5 confirm our expectation. We estimate the treatments' effects to increase (1) the probability that a subject will make a morning trip on any given weekday, (2) the total number of trips a subject makes on all weekdays in any given week, and (3) the total number of trips made by all subjects in either CG, ET, or LT on any given weekday. The coefficient estimates for $During_{ij} \times Trt_i$ and $Post_{ij} \times Trt_i$ measure the extensive margin during promotion and postpromotion, respectively. These estimates are economically negligible and almost always statistically insignificant.

In the absence of treatment-induced trips, we compute the intensive margin by multiplying the number of trips ET and LT made during promotion and postpromotion by the estimated shift in percentages shown in Table 3. Table 6 presents the results for this calculation, which are in proportion to the treatment-effect estimates in Table 3. Because this calculation does not adjust for attrition—and hence the full sample should have made more trips and more shifts than what we observe—the intensive margin shown is a conservative estimate.

According to Table 6, the treatments shifted 271 and 553 trips out of the 8:15 A.M.–9:00 A.M. window (summing columns (4) and (5)) during and after the promotional period. Over the 10-week promotion, we paid S$1,346 (S$417 to ET and S$929 to LT) to subjects for their promotion-eligible trips. Restricting the treatment effects to the during-promotion period, the average cost per shift is S$4.97 (S$1,346/271). This cost

Table 5. The Treatments Hardly Induced Additional Trips

Variables	(1) Daily individual travel probability	(2) Weekly individual total	(3) Daily total by treatment
During	−0.039	-0.196^{*}	-6.877^{***}
	(0.024)	(0.117)	(2.242)
During × *Trt*	−0.013	−0.061	−4.356
	(0.027)	(0.136)	(2.696)
Post	-0.033^{*}	-0.200^{**}	-7.271^{***}
	(0.019)	(0.098)	(1.786)
Post × *Trt*	−0.025	−0.114	-4.963^{**}
	(0.024)	(0.121)	(2.152)
No. of active subjects			−0.015
			(0.010)
No. of active subjects same trt			0.839^{***}
			(0.089)
Observations	75,981	15,329	875
R-squared	0.147	0.275	0.581
Individual fixed effects	Yes	Yes	No
Day of week fixed effects	Yes	No	Yes

Notes. The dependent variable in column (1) is a binary indicator for whether a subject makes a morning peak trip on any given weekday. The unit of observation is subject-day. The dependent variable in column (2) is the total number of trips a subject makes on all weekdays in any given week. The unit of observation is subject-week. In columns (1) and (2), the sample period for each subject is restricted to between his or her first date and last date of observation. The dependent variable in column (3) is the number of trips made by all subjects in CG, ET, or LT on any given weekday. The unit of observation is treatment group-day. Number of active subjects is defined as the number of subjects who have yet to appear for the last time in the data. Robust standard errors are in parentheses (clustered at the individual level for (1) and (2)).
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

reduces to S$1.63 when we consider the 10-month period during and after the promotion. Given that a currently in-service and on-order S$12–13 million Kawasaki Heavy Industries C151 train has a full capacity of 2,000 passengers, continuously running a fare promotion like ours that targets at 2,000 daily shifts for 3.3 years costs as much as purchasing a train and provides similar congestion relief.[9] This implies that, relying only on the contemporaneous effect and not considering trains' operating costs, the promotion is not cheaper than adding trains on a four-year horizon. However, conducting the promotion for 10 weeks and discontinuing it for the next seven months extends the breakeven point to beyond 10 years. Because the treatment effect's postpromotion decay is small, we believe that the potential to improve the intervention's cost efficiency depends on its long-term effects.

3.3. Who Shifted and Who Persisted?

In addition to the trip-level difference-in-differences analysis in model (1), we investigate at the subject level to search for factors that facilitate subjects in making desirable shifts and persisting with them. In Table 7's regressions (1)–(3), we regress each subject's prepromotion and during-promotion differences in (1) exit time, (2) percentage of trips ending before 8:00 a.m., and (3) percentage of trips ending between 8:30 a.m. and 9:00 a.m. on a variety of prepromotion travel patterns and their interactions with the dummy indicating whether the subject in question is from one of the

Table 6. Average Number of Trips Shifted

No. of trips shifted	(1) Before 7:45	(2) Before 8:00	(3) Before 8:15	(4) Bet. 8:15 and 8:30	(5) Bet. 8:30 and 9:00	(6) Bet. 9:00 and 9:30
During promotion	129^{**}	355^{***}	271^{**}	-252^{**}	−19	−6
Postpromotion	298^{**}	567^{**}	397	269	-822^{**}	156

Notes. Commuters in ET and LT combined made 6,456 and 14,406 morning trips during and after the promotion, respectively. The numbers of trips shifted are obtained by multiplying the total number of trips by the estimated percentages of shifts in Table 3.
Significance levels for the original estimates in Table 3 are indicated by $^{***}(p < 0.01)$ and $^{**}(p < 0.05)$.

Table 7. Who Made The Shifts: During Promotion and Postpromotion

Variables × *Trt*	(1) Pre-dur exit time	(2) % before 8:00	(3) % 8:30–9:00	(4) Pre-post exit time	(5) % before 8:00	(6) % 8:30–9:00
Pre- % Trips w/bus trans.	2.288	−0.021	0.015	2.518	−0.104*	0.069
	(3.257)	(0.053)	(0.057)	(4.328)	(0.057)	(0.074)
Pre- Avg. exit time (min)	−0.664	0.009	0.016*	−0.090	0.009	0.015
	(0.473)	(0.008)	(0.008)	(0.649)	(0.009)	(0.011)
Pre- Avg. subway fare	1.732	−0.012	0.012	0.170	−0.013	−0.029
	(2.922)	(0.047)	(0.051)	(3.880)	(0.051)	(0.067)
Pre- % Ending by 8:00	−11.090	0.058	0.367	1.760	0.008	0.392
	(14.723)	(0.237)	(0.257)	(19.497)	(0.258)	(0.335)
Pre- % Ending by 8:15	−39.625	0.469	1.195**	−1.310	0.465	1.042
	(33.076)	(0.533)	(0.578)	(45.614)	(0.603)	(0.784)
Pre- % Ending 8:15–8:30	−29.840	0.392	1.009**	−9.101	0.494	0.957
	(27.004)	(0.436)	(0.472)	(37.045)	(0.490)	(0.636)
Pre- % Ending 8:30–9:00	−19.087	0.163	0.589**	−2.025	0.171	0.569
	(16.593)	(0.268)	(0.290)	(23.203)	(0.307)	(0.399)
Pre- Daily avg. # trips	−16.900*	0.249*	0.012	−9.638	0.349**	−0.179
	(8.967)	(0.145)	(0.157)	(12.700)	(0.168)	(0.218)
Pre- Variability	0.207	−0.004	0.001	0.122	0.004	−0.003
	(0.203)	(0.003)	(0.004)	(0.282)	(0.004)	(0.005)
Observations	347	347	347	287	287	287
R-squared	0.073	0.117	0.111	0.065	0.089	0.168
Variables not as interacted	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Displayed are coefficient estimates for variables interacting with treatment dummy. All noninteracted variables are also included in the regression. "Pre-" means prepromotion. Variability is defined as the standard deviation of exit time (in minutes). Standard errors are in parentheses.

$^{**}p < 0.05$; $^{*}p < 0.1$.

two treatment groups (Trt_i). In regressions (4)–(6), we examine each subject's pre- and postpromotion differences in the same three measures. Prepromotion travel frequency seems to negatively affect the treatment effect of making a trip more likely to end before 8:00 A.M. However, across the board, none of prepromotion travel patterns appears to have statistically significant correlations with the treatment effects of both reducing peak-time trips and increasing prepeak travel. Although this statistical insignificance may be owing to a small sample size, the exercise leaves us no evidence that any particular prepromotion travel pattern explains the shifts or mediates the treatments' effects. While an optimist may conclude that monetary incentives work regardless of a subject's exiting travel pattern, we conservatively hypothesize that other unobserved variables may mediate the treatments' effects. This is for future research.

A few qualitative findings also provide some insights. In our exit survey, commuters cited (1) direct monetary incentives (61% "somewhat agree" or "completely agree" on "saving in fare"), (2) prestige effect (57% on promotion being "only extended to a small group"), (3) novelty effect (47% on promotion being "only offered for a limited time"), (4) alignment with intrinsic motivation (41% on having had "the plan/thought to travel earlier"), and (5) less crowded prepeak travel experience (58%) as reasons why they made the shifts during the promotion.

When asked in our focus-group discussions about their reasons for continuing to travel early after the promotion ended, participants mentioned learning about the more comfortable prepeak travel experience and forming an early travel habit. Other explanations may well exist. For instance, by conducting the experiment, we may have promoted early travel as a welcome social norm, thereby increasing its inherent value. It is also possible that some subjects responded to the incentives by making costly lifestyle adjustments and were not incentivized to incur these costs again by switching back after the promotion ended, which led to inertia—and, in turn, persistence in travel times.

Either subjects were gradually making early travel a "habitual behavior"—defined, in the sense of Becker and Murphy's (1988) theory, as traveling at an early time, similar to that of previous trips positive impact on utility—or they were making costly lifestyle changes. In either case, travel regularity and consistency appear to be important. Psychology studies, such as those by Lally et al. (2010) and Lally and Gardner (2013), argue that "to form a habit, a behavior must be carried out repeatedly in the presence of the same contextual cues" (Lally and Gardner 2013, p. S143). Since continuation and repetition are important, Borland (2010)

Table 8. The Importance of Consistence

Variables	(1) <5D gap before 8:00	(2) ≥5D gap before 8:00	(3) <5D gap bet. 8:30 and 9:00	(4) ≥5D gap bet. 8:30 and 9:00
During	0.014	0.018	0.048	−0.026
	(0.011)	(0.014)	(0.044)	(0.019)
During × *Trt*	0.095**	0.043**	−0.131**	0.030
	(0.042)	(0.020)	(0.052)	(0.024)
Post	−0.013	−0.005	0.135**	0.029
	(0.015)	(0.013)	(0.055)	(0.023)
Post × *Trt*	0.089**	0.026	−0.204***	−0.028
	(0.040)	(0.019)	(0.067)	(0.028)
Observations	9,792	40,391	9,792	40,391
R-squared	0.455	0.337	0.523	0.392
Individual fixed effects	Yes	Yes	Yes	Yes
Trip-level controls	Yes	Yes	Yes	Yes

Notes. Each column uses a subsample of the subjects either not having or having a gap of at least five consecutive calendar days (5D) during the promotional period. Trip-level controls include the before-discount trip fare, a binary indicator for whether the trip is connected to a bus transfer, and a dummy for day of week when the trip is made. Robust standard errors are in parentheses (clustered at the individual level).
$^{***}p < 0.01$; $^{**}p < 0.05$.

suggests that postintervention factors that may encourage or discourage continuation and repetition are crucial for persistence. Following these studies, we identify "gaps" in commuters' travel history—which are defined as several consecutive days during which a commuter has no travel record at the treatment stations before he or she reappears—as a variable that may disrupt travel consistency. These gaps are possibly caused by commuters' absence from work and are likely exogenous to the promotional fare. Therefore, we hypothesize that it is more difficult for commuters with a significant gap during the promotional period to form an early travel habit. Postpromotion persistence, therefore, would be weaker.

Table 8 reports the subsample analysis of two groups of commuters: those who have more than five consecutive calendar days of gaps during the promotional period and those who do not. In terms of increasing early travel and reducing peak-window trips, we find much stronger treatment effects, both during and after the promotional period, for commuters without five or more consecutive days of gaps than for those with. Notably, for a trip made by those without such gaps, the probability that it will end between 8:30 a.m. and 9:00 a.m. drops by about 20.4% postpromotion. We also examine gaps defined as 6 or more, 7 or more,..., 15 or more consecutive calendar days as a robustness check. Although separating subjects by the five-day definition yields the starkest contrast between those with and without gaps, the postpromotion reduction in peak-time trips is larger and more often statistically significant if a subject is without a gap than with—no matter how we define the gap. We include these results in the online appendix.

A potential caveat of this analysis is that the subsample division may introduce selection bias. If commuters either with or without gaps are otherwise the same, we would expect gap windows to occur randomly in commuters' travel records. Indeed, it turns out that 85% (50 of 59) of those without gaps during the promotion had such gaps during the 10 weeks before the promotion, while of all those 289 subjects who had gaps during the promotion, 53 did not have gaps during the 10 weeks before the promotion. Table 9 also shows that these two subsamples are largely balanced on prepromotion travel patterns and demographics. However, a slightly higher childcare responsibility may imply that those with gaps find it more difficult to adjust their travel schedules.

3.4. Robustness Checks

We conduct a large set of robustness checks to validate our findings. First, to address the concern about attrition, we repeat the estimation of model (1) using shrinking subsamples of subjects who had travel records in the first 1 month, 2 months,..., 7 months after the end of promotion and those who had travel records in the last week of the observation window (labeled as the "end" group). Similar to Figure 4, we plot the coefficient estimates for $During_{ij} \times Trt_i$ and $Post_{ij} \times Trt_i$ and their 95% confidence intervals for each subsample in Figure 5.

Across the board, we find no evidence that any particular cohort has an uncommon magnitude of treatment effects or a treatment effect notably different from the full sample. This is consistent with our conjecture that attrition happens randomly.

Second, to account for potential selection bias introduced by our recruitment process, we collect travel records that cover the same 60-week sample period from 200 random and anonymous commuters—100 from each treatment station—who met the inclusion

Table 9. Subsample Balance: Without ≥5 Gap Days vs. With

Prepromotion travel pattern and demographics	(1) <5D gap	(2) ≥5D gap
Personal average exit time	8:29	8:28
Personal average probability of ending trip by 7:45 A.M. (%)	1.36	3.65***
Personal average probability of ending trip by 8:00 A.M. (%)	9.17	8.78
Personal average probability of ending trip by 8:15 A.M. (%)	26.67	26.78
Personal average probability of ending trip 8:15 A.M.–8:30 A.M. (%)	25.29	26.45
Personal average probability of ending trip 8:30 A.M.–9:00 A.M. (%)	39.51	35.65
Personal average probability of ending trip 8:15 A.M.–9:00 A.M. (%)	64.80	62.10
Average weekly trips per person	4.15	3.78***
Percentage in early treatment+	38.98	33.56
Percentage in late treatment+	30.51	31.49
Percentage in control group+	30.51	34.95
Had at least one trip before 8 A.M. (%)	54.24	53.98
Average of exit time standard deviation	11.08	13.58**
Average age	40.18	35.38**
Percentage female	53.7	64.6
Income (median)+	3k–4k	3k–4k
Escort school child(ren) at least once per week + (%)	9.76	14.29
Escort school child(ren) 5 times per week + (%)	4.88	10.86
Attrition rate (%)	50.8	47.1

Note. Shown are t-test results for equal means except for statistics that are marked by +.
$^{***}p < 0.01$; $^{**}p < 0.05$.

Figure 5. (Color online) During-Promotion and Postpromotion Treatment Effects, by Cohorts Differing in Length of Postpromotion Travel Records

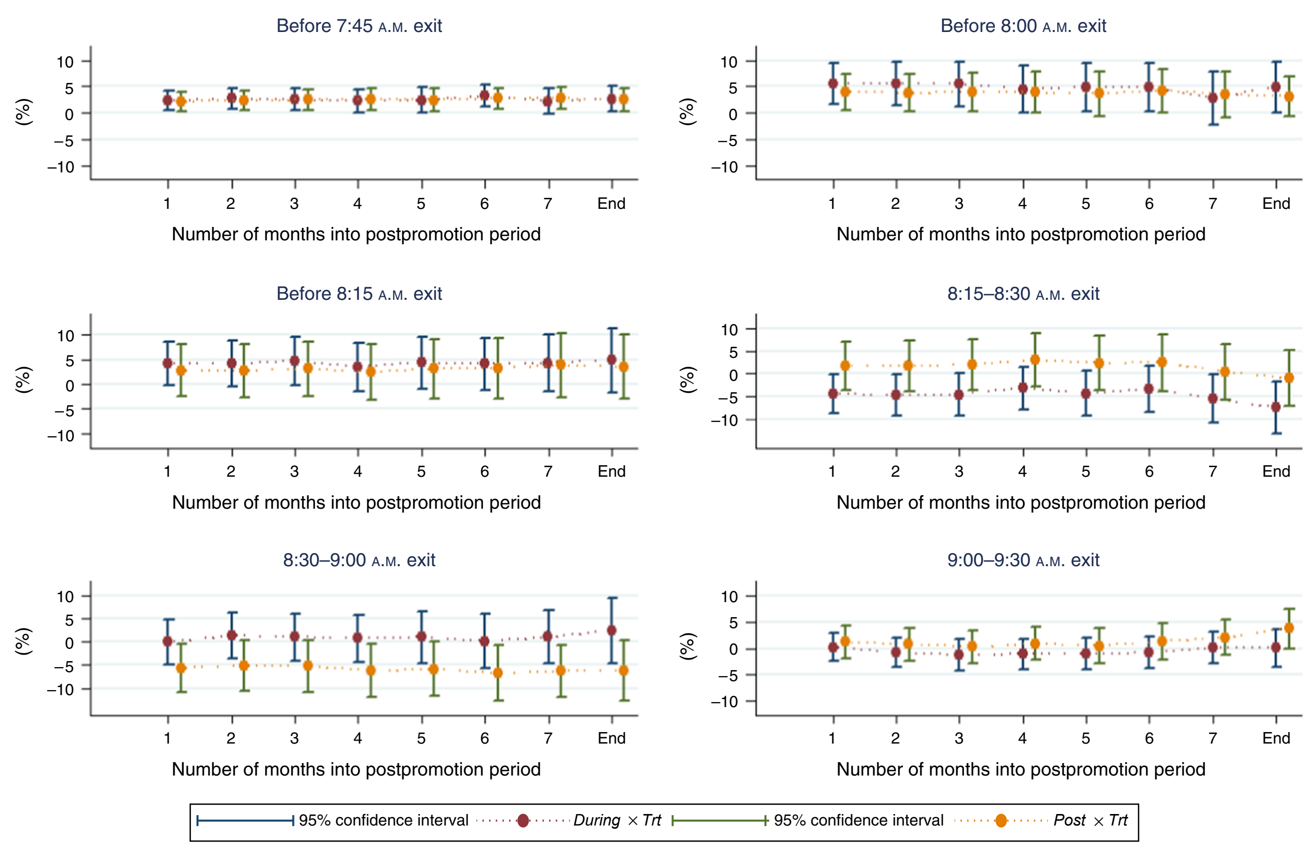


Table 10. Trip-Level Difference-in-Differences Estimates: Using Nonrecruited Commuters as Baseline

Variables	(1) Exit time (min)	(2) Before 7:45	(3) Before 8:00	(4) Before 8:15	(5) Bet. 8:15 and 8:30	(6) Bet. 8:30 and 9:00	(7) Bet. 9:00 and 9:30
During	−1.257**	0.014**	0.018*	0.015	−0.008	−0.009	0.001
	(0.638)	(0.005)	(0.009)	(0.012)	(0.011)	(0.011)	(0.006)
During × *CG*	0.096	−0.014*	−0.001	−0.002	0.020	−0.004	−0.013
	(0.992)	(0.008)	(0.015)	(0.019)	(0.020)	(0.021)	(0.013)
During × *Trt*	−1.752*	0.006	0.054***	0.041**	−0.020	−0.008	−0.013
	(0.937)	(0.008)	(0.017)	(0.019)	(0.016)	(0.017)	(0.009)
Post	0.116	0.003	−0.015	−0.021	0.016	0.015	−0.009
	(0.690)	(0.005)	(0.009)	(0.015)	(0.014)	(0.013)	(0.008)
Post × *CG*	0.831	−0.006	0.011	0.010	−0.040	0.025	0.004
	(1.151)	(0.008)	(0.015)	(0.026)	(0.026)	(0.025)	(0.015)
Post × *Trt*	−1.252	0.013	0.049***	0.035	−0.020	−0.030	0.015
	(1.090)	(0.009)	(0.016)	(0.022)	(0.020)	(0.020)	(0.012)
Observations	93,741	93,741	93,741	93,741	93,741	93,741	93,741
R-squared	0.551	0.269	0.349	0.513	0.320	0.431	0.327
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trip-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The baseline group are 200 random commuters who did not sign up to participate in the experiment but (1) met the inclusion criterion and (2) had travel records during the recruitment and promotion period. Trip-level controls include the before-discount trip fare, a binary indicator for whether the trip is connected to a bus transfer, and a dummy for day of week when the trip is made. Robust standard errors are in parentheses (clustered at the individual level).

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

criterion for the subjects in our experiment and had travel records during the recruitment and promotion period. With this additional data, we estimate the following model:

$$\begin{aligned} dv_{ij} = {} & \beta_1 During_{ij} + \beta_2 During_{ij} \times CG_i + \beta_3 During_{ij} \times Trt_i \\ & + \beta_4 Post_{ij} + \beta_5 Post_{ij} \times CG_i + \beta_6 Post_{ij} \times Trt_i \\ & + \beta_7 X_{ij} + \nu_i + \varepsilon_{ij}, \end{aligned} \tag{3}$$

which differs from the main difference-in-differences model (1) by using the nonrecruited sample as the baseline. The binary variable CG_i has a value of 1 when subject i is in CG and 0 otherwise. Estimates for β_2 and β_5 measure the difference between the nonrecruited sample and CG during promotion and postpromotion. The estimation results in Table 10 show that the estimates for β_2 and β_5 are small and statistically insignificant at almost all times. The treatment effect estimates β_3 and β_6 are quantitatively similar to their counterparts in Table 3. This exercise provides assurance that the selection bias is small and strengthens the external validity of our field experiment.

Finally, in the online appendix, we replicate Tables 5, 8, and 10 by treating ET and LT separately. We also replicate Tables 7 and 8 using dependent variables that are among the seven used in model (1) but, because of space limits, do not appear in the original tables. The main points made by all of these tables remain qualitatively unchanged.

4. Conclusion

We present findings from a field experiment showing that offering free prepeak travel helps to smooth public-transit demand during morning peak times. The effect persists seven months after the financial incentives' discontinuation. Previous studies have shown that whether financial incentive alters behavior in the intended way is a context-specific issue, and ours is no exception. Discovering the behavioral mechanisms that drive the results would contribute substantially to our understanding, even in contexts other than urban transit. We have made some attempts, but we have a long way to go before understanding the complex issue of why persistence (does not) happen.

In our qualitative survey, many commuters said that they were intrinsically motivated to avoid the crowded peak hours and our promotion simply provided additional incentives. This alignment of interests seems to have made our job easier. However, an early study by Deci (1971) and its many follow-ups on self-determination theory argue that extrinsic rewards often undermine intrinsic motivation and impair autonomy, which has an adverse effect on intended outcomes. We do not know whether such crowding out occurs in our study or, if it does, to what extent.

Apart from underexplored behavioral mechanisms, we note that the cost effectiveness of our promotional fare scheme appears to leave room for further improvement. The estimated cost of shifting a trip out of the peak window is S$4.97, while the subway fare is about S$2. This suggests that a large portion of the expenses in fact went to subsidize trips that would have occurred before the peak window without any incentives. One direction for improvement would be to tailor incentive schemes based on individual travel patterns. For instance, we could offer rewards that are collectible only if commuters make shifts away from their own previous exit times. In this way, we could avoid paying for early trips that would have occurred in any case. Travel-card and automated fare-collection systems already render individualized incentive schemes of this kind technologically feasible. More studies are warranted on the design, effectiveness, and ethics of such schemes.

Acknowledgments
Mai Aijing, Poh Yong Fei, Pham Minh Trang, and a group of National University of Singapore students provided outstanding research assistance in conducting and analyzing the experiment. The authors thank the department editor, two anonymous reviewers, and seminar participants on various occasions for helpful comments that significantly improved the paper. This research reflects the views of the authors and does not represent the official position of the Ministry of Transport, Singapore. All remaining errors are the authors'.

Endnotes
[1] See http://www.healthhub.sg/programmes/37/nsc (accessed January 1, 2016).

[2] See Ho et al. (2018) for an example on private vehicles.

[3] According to Land Transport Authority (2015).

[4] For Hong Kong, see http://www.mtr.com.hk/en/customer/main/early_bird.html (accessed January 1, 2016), and for Beijing, see http://usa.chinadaily.com.cn/china/2015-12/22/content_22778147.htm (accessed January 1, 2016).

[5] This number is based on travel records stored in the central database.

[6] Exceptions include concession cards and those combined with a credit card.

[7] The promotion ended on November 28, the beginning of a school holiday, which may partially explain the delays in exit time and the reduction in the percentage of trips before 8:00 A.M. Note that the exit time is also delayed and the percentage reduced for CG in the weeks immediately after the promotion's end.

[8] The new trains on order are estimated to cost S$12 to S$13 million each, according to Tan (2015).

[9] A brief note on the calculation: Shifting 2,000 trips costs S$9,920 daily (S$4.97 × 2,000). So S$12 million will last 1,210 days, or 3.3 years.

References

Acland D, Levy MR (2015) Naiveté, projection bias, and habit formation in gym attendance. *Management Sci.* 61(1):146–160.

Allcott H, Mullainathan S (2010) Behavioral science and energy policy. *Science* 327(5970):1204–1205.

Anderson ML (2014) Subways, strikes, and slowdowns: The impacts of public transit on traffic congestion. *Amer. Econom. Rev.* 104(9):2763–2796.

Basso LJ, Silva HE (2014) Efficiency and substitutability of transit subsidies and other urban transport policies. *Amer. Econom. J. Econom. Policy* 6(4):1–33.

Basso LJ, Guevara CA, Gschwender A, Fuster M (2011) Congestion pricing, transit subsidies and dedicated bus lanes: Efficient and practical solutions to congestion. *Transport Policy* 18(5):676–684.

Becker GS, Murphy KM (1988) A theory of rational addiction. *J. Political Econom.* 96(4):675–700.

Bertrand M, Duflo E, Mullainathan S (2004) How much should we trust differences-in-differences estimates? *Quart. J. Econom.* 119(1):249–275.

Borland R (2010) Habits and temporality: A commentary on Hall and Fong's temporal self-regulation theory. *Health Psych. Rev.* 4(2):66–69.

Cervero R (1986) Time-of-day transit pricing: Comparative us and international experiences. *Transport Rev.* 6(4):347–364.

Charness G, Gneezy U (2009) Incentives to exercise. *Econometrica* 77(3):909–931.

De Witte A, Macharis C, Lannoy P, Polain C, Steenberghen T, Van de Walle S (2006) The impact of "free" public transport: The case of Brussels. *Transportation Res. Part A: Policy Practice* 40(8):671–689.

Deci EL (1971) Effects of externally mediated rewards on intrinsic motivation. *J. Personality Soc. Psych.* 18(1):105–115.

Fujii S, Kitamura R (2003) What does a one-month free bus ticket do to habitual drivers? An experimental analysis of habit and attitude change. *Transportation* 30(1):81–95.

Gneezy U, Meier S, Rey-Biel P (2011) When and why incentives (don't) work to modify behavior. *J. Econom. Perspect.* 25(4):191–210.

Henn L, Karpouzis G, Sloan K (2010) A review of policy and economic instruments for peak demand management in commuter rail. *Proc. 33rd Australasian Transport Res. Forum Conf., Canberra.*

Ho T-H, Png IPL, Reza S (2018) Sunk cost fallacy in driving the world's costliest cars. *Management Sci.* 64(4):1761–1778.

Lally P, Gardner B (2013) Promoting habit formation. *Health Psych. Rev.* 7(Suppl. 1):S137–S158.

Lally P, Van Jaarsveld CH, Potts HW, Wardle J (2010) How are habits formed: Modelling habit formation in the real world. *Eur. J. Soc. Psych.* 40(6):998–1009.

Land Transport Authority (2013) Travel demand management. Report, Land Transport Authority of Singapore, Singapore. https://www.lta.gov.sg/content/dam/ltaweb/corp/PublicTransport/files/FAQ-free-travel-20130415.pdf.

Land Transport Authority (2015) Free pre-peak travel extended until 30 June 2016. News release (May 26), Land Transport Authority of Singapore, Singapore. http://healthaffairs.org/blog/2015/05/13/the-aca-the-ada-and-wellness-program-incentives/.

Loewenstein G, Price J, Volpp K (2016) Habit formation in children: Evidence from incentives for healthy eating. *J. Health Econom.* 45(January):47–54.

Madison K (2015) The ACA, the ADA, and wellness program incentives. *Health Affairs Blog* (May 13), http://healthaffairs.org/blog/2015/05/13/the-aca-the-ada-and-wellness-program-incentives/.

Paulley N, Balcombe R, Mackett R, Titheridge H, Preston J, Wardman M, Shires J, White P (2006) The demand for public transport: The effects of fares, quality of service, income and car ownership. *Transport Policy* 13(4):295–306.

Royer H, Stehr MF, Sydnor JR (2015) Incentives, commitments and habit formation in exercise: Evidence from a field experiment with workers at a Fortune-500 company. *Amer. Econom. J. Appl. Econom.* 7(3):51–84.

Sharaby N, Shiftan Y (2012) The impact of fare integration on travel behavior and transit ridership. *Transport Policy* 21(May):63–70.

Tan C (2015) First of 45 new trains for SMRT arrives. *Straits Times* (May 21), http://www.straitstimes.com/singapore/transport/first-of-45-new-trains-for-smrt-arrives.

Thøgersen J (2009) Promoting public transport as a subscription service: Effects of a free month travel card. *Transport Policy* 16(6):335–343.